Privacy-Preserving Face Anonymization via Frequency-Aware Structured Perturbation

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***Abstract*—While face recognition systems have become deeply integrated into everyday infrastructure, they present an ongoing challenge to individual privacy. Images that look entirely normal to human viewers often still generate highly discriminative identity features in automated pipelines. Current attempts to anonymize these images—such as GAN-based identity swapping or standard adversarial perturbations—typically struggle to balance visual realism, computational efficiency, and robust cross-model privacy. In this paper, we propose *Frequency-Aware Structured Perturbation* (FASP), an adversarial cloaking framework designed to navigate these tradeoffs more effectively. FASP uses an edge-sensitive Laplacian mask and a multi-frequency sinusoidal generator to dynamically restrict perturbation energy to the high-frequency regions of an image. Because human perception is less sensitive to alterations in these contour-heavy areas, FASP can suppress identity embeddings without distorting the global semantic structure of the face. We benchmark this approach against PrIdentity (an adaptive L_p baseline) and PRISM (an exploratory wavelet-based geometric framework). Across LFW, CelebA, and CelebA-HQ datasets, and evaluated against ArcFace, FaceNet/CASIA, VGGFace2, and MobileFace, our results demonstrate that FASP achieves 100% Rank-1 anonymization across multiple gallery configurations. Furthermore, it yields an SSIM of up to 0.9238 while minimizing geometric distortion to a bounding-box shift of 1.55 pixels. Through these results, we establish the practical value of multi-frequency sinusoidal priors and edge-guided perturbation masking for robust, interpretable facial anonymization.**

***Index Terms*—face anonymization, adversarial perturbation, frequency-aware cloaking, privacy-preserving machine learning, identity suppression, Laplacian edge masking, sinusoidal perturbation, privacy–utility tradeoff, biometric privacy, transferable adversarial attacks**

I. Introduction

Face recognition technology has expanded far beyond specialized biometric applications, becoming a standard component of consumer devices, social media platforms, and public surveillance networks. This shift fundamentally alters individual privacy expectations. A single photograph can now be evaluated by dozens of automated models without the subject's consent or awareness. More critically, an image that looks completely benign to a human viewer can still harbor subtle identity cues that machine learning pipelines exploit. This divergence between human and machine perception forms the core challenge we address in this study.

Modern recognition networks—such as ArcFace [4] or FaceNet [3]—encode facial identity into high-dimensional embeddings that remain remarkably robust to normal visual distortions. While simple visual redaction techniques like blurring or pixelation might hide a face from human eyes, they frequently fail against automated systems, especially when pre-processing techniques like super-resolution are applied.

To achieve more robust machine-level anonymization, researchers have increasingly turned to adversarial perturbations. By calculating imperceptible but targeted modifications to an image, adversarial cloaking shifts the underlying recognition embedding away from its true identity. Approaches like LowKey [10], TIP-IM [11], and DIM/MT-DIM [17] demonstrate that these attacks can transfer across different proprietary models. Yet, most of these techniques rely on fixed perturbation limits applied uniformly across the image, largely ignoring spatial and frequency-domain characteristics. Recent work such as PrIdentity [1] addresses part of this gap via an adaptive L_p regularization scheme that adjusts the norm exponent spatially, but it still operates predominantly in the pixel domain.

We approach this problem by explicitly recognizing that the structure of an adversarial perturbation matters just as much as its magnitude. Human vision tolerates high-frequency noise relatively well—particularly around edges, textures, and contour transitions—while recognition models rely heavily on these exact features for discrimination. By exploiting this asymmetry, we can achieve significantly better identity suppression without increasing the perceptual cost. Our framework, *Frequency-Aware Structured Perturbation* (FASP), replaces free-form noise with a constrained subspace. Using a Laplacian edge attention map paired with a multi-frequency sinusoidal carrier, FASP spatially localizes and spectrally structures its perturbations.

Rather than providing a standard list of disjointed contributions, we frame our methodology around three interconnected technical advances. First, we introduce a frequency-aware masking strategy that gates perturbation energy strictly toward high-frequency facial contours. Second, we transition away from random noise initialization by employing a multi-frequency sinusoidal synthesis module, providing a more interpretable and controllable adversarial prior. Finally, we integrate these structural priors into a joint privacy–utility optimization objective that balances a softplus margin loss for identity suppression against LPIPS and MSE penalties.

Through extensive evaluation on LFW, CelebA, and CelebA-HQ, we demonstrate how this frequency-focused approach translates into concrete improvements over pixel-domain baselines.

The paper proceeds by reviewing relevant literature in Section II and detailing the PrIdentity baseline in Section III. We present the FASP methodology in Section IV, followed by our experimental setup and primary results in Sections V and VI. Section VII discusses an exploratory Riemannian extension, PRISM, while Sections VIII and IX contextualize our findings and their broader ethical implications. We conclude in Section XI.

II. Related Work

Prior literature on face anonymization generally falls into a few distinct categories: generative synthesis, adversarial cloaking, frequency-domain modifications, and transferability-focused attacks.

A. GAN-Based and Synthesis Methods

Several researchers have proposed replacing detected faces entirely. CIAGAN [8], for example, uses a conditional GAN to substitute a subject's identity while attempting to maintain the original pose and expression. DeepPrivacy [12] explores a similar path, conditioning its generation on background context rather than explicit identity matching. Subsequent systems like G2Face and RiDDLE [9] further refine synthesis fidelity, with RiDDLE leveraging a reversible de-identification framework within the latent space. More recently, Diff-Privacy [13] applies diffusion priors to achieve highly realistic anonymized outputs. Despite these advances in visual quality, generative approaches share a fundamental vulnerability: they frequently alter non-identity attributes like lighting, accessories, or background statistics, and can be computationally prohibitive for real-time deployment.

B. Adversarial Cloaking

Adversarial cloaking targets the recognition embedding directly rather than synthesizing a new face from scratch. LowKey [10] optimizes pixel-space perturbations specifically to defeat commercial social media filters, showing that imperceptible noise can severely degrade verification performance. To improve how well these perturbations transfer to unknown models, TIP-IM [11] introduces targeted identity masks, while DIM and MT-DIM [17] incorporate input diversity during optimization. FIT adapts the perturbation to facial identity geometry, while FALCO extends this line by jointly optimizing privacy and visual fidelity in an end-to-end framework. PrIdentity [1] currently represents a strong state-of-the-art baseline in this category. Instead of using a fixed global budget, it relies on an adaptive L_p norm that shifts between sparse and smooth perturbation geometries depending on the local image context.

C. Frequency-Domain Approaches

While the adversarial robustness literature frequently highlights the spectral sensitivities of convolutional networks—noting that mid-to-high spatial frequencies carry the bulk of texture and contour information—most anonymization techniques ignore this property. Frequency-targeted attacks outside the privacy domain have long exploited this sensitivity by concentrating energy where it is most spectrally disruptive. Yet, practical face anonymization systems continue to operate primarily in the pixel domain, leaving the perceptual advantages of frequency modulation largely unexploited.

D. Research Gaps

As outlined in Table I, the current ecosystem of face anonymization forces a recurring compromise between computational expense, perceptual realism, and cross-model robustness. Generative pipelines produce convincing visuals but struggle with efficiency and black-box transferability. Standard adversarial attacks are fast, but their uniform perturbation budgets often degrade perceptually smooth areas of the face unnecessarily. While PrIdentity mitigates this via adaptive norms, it remains inherently a pixel-domain approach. FASP bridges this divide by embedding explicit frequency-domain constraints directly into both the synthesis of the perturbation and its spatial distribution.

III. Baseline Method: PrIdentity

Because FASP is explicitly designed to improve upon pixel-domain adaptive methods, we evaluate it primarily against the PrIdentity framework [1]. PrIdentity formulates face anonymization as a constrained optimization problem, aiming to maximize identity divergence while penalizing visual distortion through a learned L_p norm exponent.

A. Problem Formulation

Given an input face image $I \in \mathbb{R}^{H \times W \times 3}$ and a recognition model $f(\cdot)$, PrIdentity seeks an anonymized output

$$A = I + P, \tag{1}$$

where P is the learned perturbation. The objective is to produce A such that the recognition embedding of A is far from that of I, while the image remains visually close to the original under a perceptual metric.

The overall optimization objective is:

$$\min_{P} \; \mathcal{L}_{\text{priv}}(I, I + P) + \lambda \, \mathcal{L}_{\text{util}}(I, I + P), \tag{2}$$

where $\mathcal{L}_{\text{priv}}$ penalizes identity similarity, $\mathcal{L}_{\text{util}}$ penalizes visual deviation, and λ governs the privacy–utility tradeoff. This formulation is significant because it does not erase identity at arbitrary cost; rather, it learns the minimum perturbation needed to cross the recognition decision boundary for a given tolerance parameter λ.

B. Margin-Based Privacy Loss

PrIdentity expresses the privacy term as a margin-based loss over embedding distances:

$$\mathcal{L}_{\text{priv}} = \sum_{i} \max\big(0, \; \alpha - D(z_I, z_A)\big), \tag{3}$$

where $z_I = f(I)$ and $z_A = f(A)$ are the embeddings of the original and anonymized images, $D(\cdot, \cdot)$ is a dissimilarity

TABLE I: Comparison of Face Anonymization Methods

Method	Privacy Strength	Visual Quality	Generalization	Perturbation Type	Computational Cost
CIAGAN [8]	Strong	Medium–High	Medium	GAN synthesis	High
RiDDLE [9]	Medium	Medium	Medium	Latent editing	Medium
DeepPrivacy [12]	Medium	High	Medium	GAN-based	High
LowKey [10]	High	High	Good	Adversarial perturbation	Medium
TIP-IM [11]	High	High	Good	Transferable perturbation	Medium
Diff-Privacy [13]	Medium	High	Medium	Diffusion editing	High
PrIdentity [1]	High	High	Good	Adaptive L_p pixel perturbation	Medium
FASP (Ours)	**High**	**High**	**Good–Strong**	**Structured freq.-domain perturbation**	**Medium**

measure, and α is the target privacy margin. When the distance $D(z_I, z_A)$ already exceeds α, the margin term vanishes and the optimizer incurs no further privacy penalty. This behavior prevents unnecessary over-perturbation once the privacy target has been achieved—a critical property for preserving utility.

C. Adaptive L_p Regularization

The defining innovation of PrIdentity is its adaptive norm regularizer, which learns one or more norm exponents p from data rather than fixing them a priori:

$$\mathcal{L}_{\text{util}} = \left(\sum_j |P_j|^p + \epsilon \right)^{1/p}, \tag{4}$$

where P_j denotes the j-th component of the perturbation and ϵ is a small constant for numerical stability. The learned exponent p typically falls within the range $[1, 2]$: values near $p \approx 1$ encourage sparse, sharply localized perturbations akin to L_1 attacks, while values near $p \approx 2$ produce smoother, more distributed modifications analogous to standard L_2 perturbations. This adaptive behavior allows the optimizer to choose the perturbation geometry best suited to each input, a flexibility that fixed-norm baselines cannot match.

In spatial form, the adaptive regularization can also be expressed as:

$$\mathcal{L}_{\text{reg}} = \|\delta\|_{p(x)} = \left(\sum_i |\delta_i|^{p(x)} \right)^{1/p(x)}, \tag{5}$$

where $p(x)$ denotes the potentially image-dependent learned exponent.

The PrIdentity architecture is shown in Fig. 1. The pipeline takes the original face I, computes recognition loss and utility penalty, and projects the perturbation P into the adaptive norm ball at each iteration. The method operates without an explicit identity-source target, making it suitable for unsupervised anonymization without reference images.

D. Qualitative and Quantitative Results

Fig. 2 shows the qualitative before-and-after output of PrIdentity, demonstrating that the anonymized images retain the overall facial structure while embedding-space identity is suppressed. The trade-off behavior is characterized in Fig. 3, which plots SSIM against cosine similarity across a range of perturbation budgets.

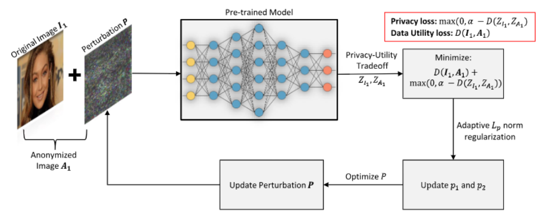


Fig. 1: PrIdentity optimization pipeline. An input face is iteratively perturbed through a recognition model, with the adaptive L_p norm projecting the update into a bounded ball at each step. The result is an anonymized output that suppresses the embedding while remaining visually close to the original.

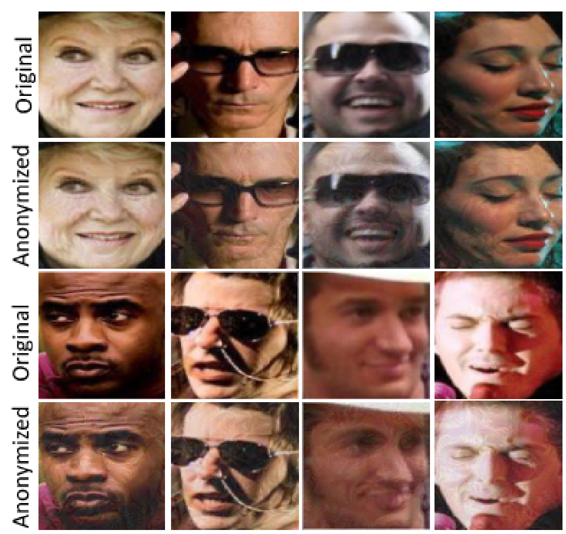


Fig. 2: PrIdentity qualitative results. Original images (left column) versus anonymized outputs (right column). Overall facial layout, expression, and color distribution are preserved while embedding-level identity cues are disrupted.

Quantitatively, PrIdentity reduces the LFW verification TPR at FPR$= 0.001$ from 0.994 (original) to 0.002 under ArcFace—a suppression factor exceeding $497\times$. On CelebA single-image anonymization, Rank-1 identification accuracy under ArcFace drops from 64.46% to 2.55%, while SSIM is maintained above 0.84. Geometric preservation, measured by

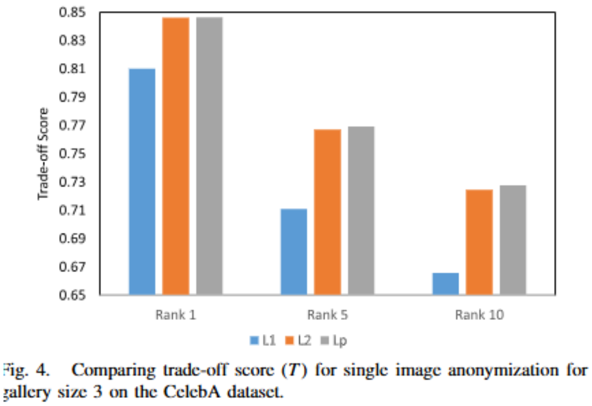


Fig. 3: PrIdentity privacy–utility tradeoff curve. Increasing the perturbation budget (controlled by α and λ) reduces cosine similarity (stronger privacy) at the cost of decreasing SSIM (lower utility). The curve defines the Pareto frontier achievable by the adaptive L_p formulation.

MTCNN bounding-box centroid displacement, reaches 2.65 pixels on CelebA-HQ, the lowest among all generative and prior perturbation-based baselines at the time of publication.

E. Strengths and Limitations

While PrIdentity successfully demonstrates the value of input-adaptive optimization without reference images, its reliance on a purely pixel-domain loss leaves significant room for improvement. The adaptive norm lacks an explicit mechanism to distinguish between smooth semantic regions (like cheeks) and high-frequency identity features (like eye contours). If the perturbation budget is not perfectly calibrated, the model either fails to anonymize the face fully or introduces visible artifacts in otherwise smooth patches. Moreover, its cross-architecture transferability remains inconsistent. FASP was developed explicitly to embed spectral awareness into the optimization loop to resolve these exact limitations.

IV. Proposed Method: FASP

A. Motivation

FASP builds upon three related observations regarding face perception. First, deep recognition models draw their discriminative power disproportionately from high-frequency spatial content; sharp edges, fine textures, and intricate contour transitions are what separate distinct identities in the embedding space. Second, human observers are notably forgiving of noise injected into these exact high-frequency regions, provided the broad low-frequency layout (such as overall face shape and skin tone) is left undisturbed. Finally, structured perturbations tend to transfer better across different model architectures than unstructured random noise. When an adversarial attack is grounded in a meaningful spatial prior, it guides the optimizer toward more robust, generalized disruptions.

While PrIdentity implicitly leverages some of these concepts through its adaptive norm, it still operates blindly in the frequency domain. FASP directly bridges this gap. By constructing a perturbation from a Laplacian edge attention map modulated by a multi-frequency sinusoidal carrier, FASP restricts adversarial noise precisely to the areas where it is most perceptually benign and computationally effective.

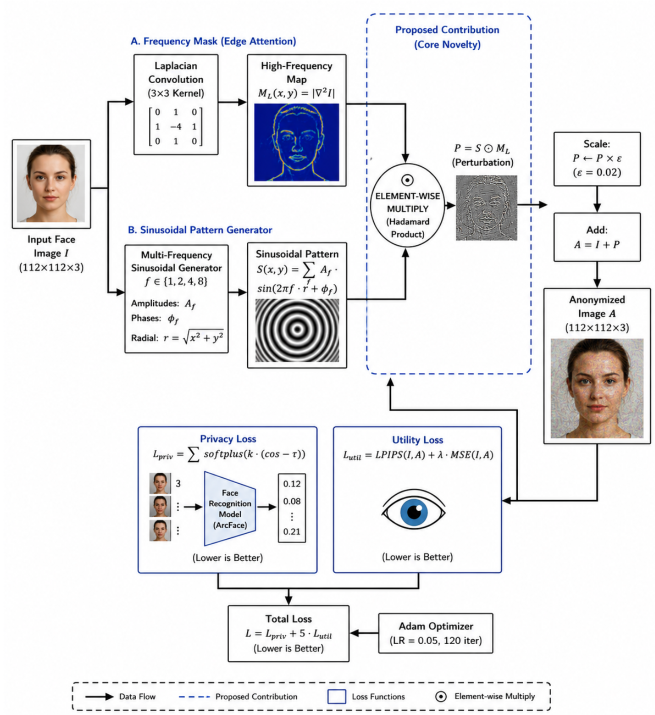


Fig. 4: FASP pipeline architecture. Starting from an input face image, the method extracts a Laplacian edge attention map, generates a multi-frequency sinusoidal carrier signal, and combines them via Hadamard product to produce a structured perturbation prior. The prior is then scaled, added to the input, and updated iteratively through a joint privacy–utility objective evaluated against a pre-trained recognition model.

B. System Overview

The FASP pipeline, detailed in Fig. 4, relies on a multi-stage process. The system extracts a Laplacian edge map from the conditioned input, normalizes it into an attention mask, and generates a multi-frequency sinusoidal carrier. These two signals are merged via a Hadamard product, scaled to meet a strict perturbation budget, and clipped to valid pixel ranges. The resulting anonymized image is then evaluated against both a privacy loss (measuring embedding similarity) and a utility loss (incorporating perceptual and pixel-wise deviations) to calculate gradients for the next optimization step.

The rationale behind our frequency masking strategy becomes clear when observing Fig. 5. Compared to uniform or random masking, our Laplacian approach safely ignores flat regions of the face, concentrating all available perturbation budget on texturally dense areas. The sinusoidal patterns generated prior to masking (shown in Fig. 6) provide a controllable and spectrally localized starting point, avoiding

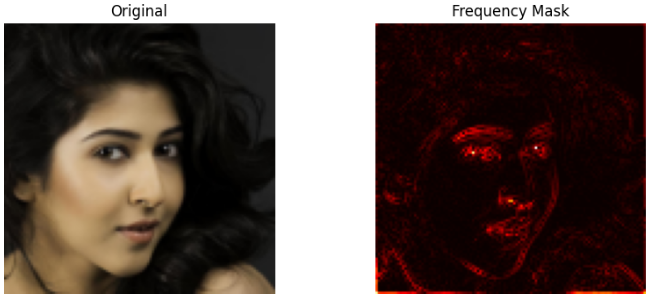


Fig. 5: Comparison of frequency masking strategies. Left: FASP edge-aware Laplacian mask, which concentrates attention on contours and texture transitions. Middle: uniform mask, which distributes perturbation budget without selectivity. Right: random mask, which wastes budget on uninformative locations. FASP's Laplacian mask localizes the perturbation at identity-sensitive regions while leaving flat areas unmodified.

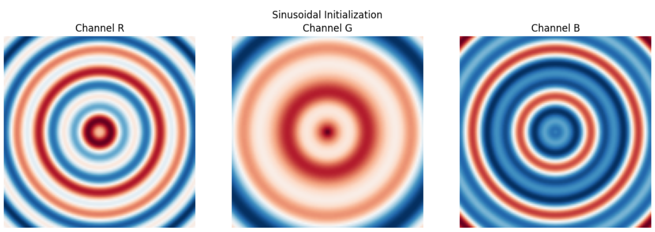


Fig. 6: Multi-frequency sinusoidal carrier patterns generated by the FASP synthesis module prior to edge-mask application. Each row corresponds to a different frequency component $f \in \mathcal{F}$. The structured oscillatory patterns provide a controllable, interpretable perturbation prior that is more spectrally specific than random noise initialization.

the chaotic frequency distribution inherent to random noise initialization.

C. Frequency Mask Extraction

To capture high-frequency details, FASP relies first on a Laplacian edge attention map. For an image $I \in \mathbb{R}^{H\times W\times 3}$, we calculate the absolute response of the spatial Laplacian:

$$M_L(x, y) = \left|\nabla^2 I(x, y)\right|, \tag{6}$$

where ∇^2 denotes the spatial Laplacian. In discrete implementation, this corresponds to convolution with the kernel

$$K_{\nabla^2} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \tag{7}$$

which computes the second-order spatial derivative at each pixel. Smooth regions of the face—cheeks, forehead, background—produce near-zero Laplacian response, whereas edges, eye contours, lip boundaries, and fine texture transitions yield strong activations. Since these high-activation locations correspond closely to the regions from which recognition models extract discriminative features, the mask provides a principled spatial allocation of the perturbation budget.

The raw Laplacian response is normalized into an edge attention map

$$A_{\text{edge}}(x, y) = \frac{M_L(x, y)}{\max_{x',y'} M_L(x', y') + \delta}, \tag{8}$$

where $\delta > 0$ is a small constant that prevents numerical instability when the image contains very little edge content. The normalized map A_{edge} takes values in $[0, 1]$, with larger values indicating regions where FASP concentrates perturbation energy.

D. Multi-Frequency Sinusoidal Synthesis

Instead of relying on isotropic random noise, FASP employs a multi-frequency sinusoidal carrier. The synthesis module generates a structured signal:

$$S(x, y) = \sum_{f\in\mathcal{F}} A_f \sin\bigl(2\pi f\, r(x, y) + \phi_f\bigr), \tag{9}$$

where

$$r(x, y) = \sqrt{(x - x_0)^2 + (y - y_0)^2} \tag{10}$$

is the radial coordinate from the patch center (x_0, y_0), $\mathcal{F}$ is the chosen set of spatial frequencies, A_f is the amplitude for frequency f, and ϕ_f is the corresponding phase offset. The radial parameterization produces concentric oscillatory patterns that naturally follow the roughly circular geometry of a face region.

This formulation provides several distinct advantages. Lower frequency components help the perturbation survive standard image compression, while higher frequencies directly attack texture features. Furthermore, because the parameter space is restricted to amplitudes and phases, the resulting perturbations remain interpretable and spectrally constrained, preventing the optimizer from descending into perceptually chaotic noise patterns.

E. Structured Perturbation Synthesis via Hadamard Product

To enforce spatial gating, FASP fuses the edge attention map and the sinusoidal carrier through a simple Hadamard product:

$$P_{\text{raw}}(x, y) = A_{\text{edge}}(x, y) \odot S(x, y). \tag{11}$$

This pixel-wise multiplication ensures that the sinusoidal oscillations only materialize where the Laplacian mask allows them. Smooth regions receive effectively zero perturbation, forcing the optimizer to utilize its perceptual budget exclusively in areas where noise blends naturally with existing image contours.

F. Perturbation Scaling and Image Anonymization

The raw perturbation is projected into the allowed perturbation budget ϵ:

$$P = \epsilon \cdot \text{clip}(P_{\text{raw}}, -1, 1), \tag{12}$$

where ϵ controls the maximum pixel-space magnitude of the perturbation. The anonymized image is then formed by adding the perturbation and clipping to the valid intensity range:

$$A = \text{clip}(I + P,\ 0,\ 1). \tag{13}$$

The clipping in Eq. (13) ensures that the anonymized image A remains a valid image tensor at every optimization step. The additive formulation in Eq. (1) keeps the pipeline fully differentiable, allowing gradient-based optimization of any loss applied to A.

G. Privacy Loss

FASP adopts a softplus-smoothed margin loss over cosine similarity to measure identity leakage:

$$\mathcal{L}_{\text{priv}} = \sum_{k} \text{softplus}\Big(\kappa \left(\cos(z_I^{(k)}, z_A^{(k)}) - \tau\right)\Big), \tag{14}$$

where $z_I^{(k)} = f_k(I)$ and $z_A^{(k)} = f_k(A)$ are the normalized embeddings under recognition model k, $\cos(\cdot, \cdot)$ denotes cosine similarity, $\tau \in [0, 1]$ is the target similarity threshold below which anonymization is considered successful, and $\kappa > 0$ sharpens the effective margin. The softplus function $\text{softplus}(u) = \ln(1 + e^u)$ acts as a smooth approximation to the hinge function: when the cosine similarity already falls below τ, the loss is exponentially small; when the similarity remains high, the loss grows approximately linearly, providing a strong gradient signal. This behavior focuses optimization effort on cases where identity is still leaking while avoiding unnecessary distortion once the privacy target is reached.

H. Utility Loss

The utility term penalizes both perceptual and pixel-level deviation from the original:

$$\mathcal{L}_{\text{util}} = \text{LPIPS}(I, A) + \lambda_{\text{mse}}\, \text{MSE}(I, A), \tag{15}$$

where LPIPS [7] is the Learned Perceptual Image Patch Similarity metric, which penalizes perceptual changes as measured by a pre-trained feature network, and λ_{mse} is a scalar weight balancing the two terms. The LPIPS term is the primary perceptual constraint: it discourages changes that are visible to the human visual system even when the pixel error is small. The MSE term provides a secondary pixel-space anchor that prevents unbounded perturbation growth in regions where the LPIPS network is not sensitive.

I. Total Objective and Optimization

The total loss that FASP minimizes is

$$\mathcal{L} = \mathcal{L}_{\text{priv}} + \lambda_u\, \mathcal{L}_{\text{util}}, \tag{16}$$

where $\lambda_u > 0$ controls the global privacy–utility tradeoff. The optimization is performed over the perturbation parameters $\theta_P = \{A_f, \phi_f\}_{f \in \mathcal{F}}$ that define the sinusoidal carrier:

$$\min_{\theta_P}\ \mathcal{L}(I,\ I + P_{\theta_P}). \tag{17}$$

Unlike PrIdentity, which optimizes over a free perturbation vector, FASP restricts the search to the structured subspace defined by the sinusoidal amplitudes and phases. This constraint reduces the effective dimensionality of the optimization and biases the solution toward perturbations with a controlled frequency profile, improving interpretability and—empirically—transferability.

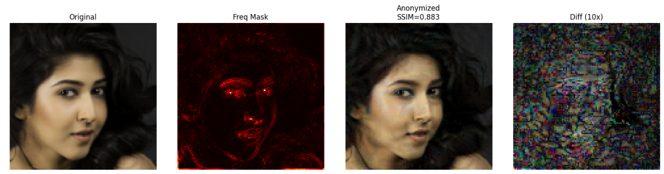


Fig. 7: Original faces (left) versus FASP anonymized outputs (right). Facial layout, expression, hair, and coarse skin color are preserved, while identity-discriminative textures around the eye region, lip boundaries, and contour transitions are disrupted. No blocking, blurring, or generative reconstruction artifacts are introduced.

The comparison between FASP's qualitative outputs and the originals is shown in Fig. 7, which demonstrates that the anonymized faces retain global structure, expression, and hair while disrupting fine texture and edge-level identity cues. Fig. **??** provides additional before-and-after comparisons across a broader range of identities, poses, and lighting conditions.

V. Experimental Setup

A. Datasets

We evaluate our framework across three established face recognition benchmarks. **LFW** (Labeled Faces in the Wild [14]) provides an unconstrained verification baseline spanning diverse real-world identities. **CelebA** [15] introduces significant variability in lighting, pose, and facial expression, which allows us to stress-test generalization. Finally, we use **CelebA-HQ** [16] to scrutinize perceptual utility; its high-resolution format exposes artifacts far more clearly than standard datasets. Prior to optimization, all images are passed through an MTCNN pipeline [6] for standard keypoint alignment and bounding box cropping, and are strictly normalized to the $[0, 1]$ range.

B. Recognition Models

To accurately characterize both white-box performance and black-box transferability, we select five distinct recognition architectures. **VGGFace2/ResNet-50** operates as our primary white-box surrogate, meaning the optimizer directly computes gradients against its embedding space. We reserve **ArcFace** [4] and **MobileFace** as strict black-box evaluation targets. **FaceNet/CASIA-WebFace** [3] serves as an intermediate target to assess how well perturbations transfer to functionally similar architectures, while **LightCNN-29** [5] provides a baseline comparison against the original PrIdentity evaluation metrics.

TABLE II: Evaluation Metrics Used in Experiments

Metric	What it Measures	Better
SSIM	Structural visual similarity	↑
PSNR	Peak signal-to-noise ratio	↑
LPIPS	Perceptual image difference	↓
Cosine similarity	Embedding identity proximity	↓
TPR @ FPR$= 0.001$	Verification recognition rate	↓
Rank-1 Accuracy	Identification recognition rate	↓
Trade-off score T	Joint privacy–utility summary	↑
Bounding-box distance	Geometric structure drift	↓

C. Implementation Details

FASP perturbation parameters $\{A_f, \phi_f\}$ are optimized using Adam with a learning rate of 10^{-2}, run for up to 100 iterations per image or batch. The perturbation budget ϵ is varied across experiments in the range of 8 to 16 pixel units (normalized), corresponding to ablation variants V0 through V4. The trade-off coefficient λ_u is swept to generate the privacy–utility Pareto curve. LPIPS is computed using the AlexNet-based reference network. PRISM experiments were conducted on a dual-GPU Kaggle T4 instance; FASP experiments require a single CUDA-capable GPU with at least 8 GB of memory.

D. Evaluation Metrics

Table II summarizes the metrics used throughout the evaluation. **SSIM** [2] measures structural similarity between the original and anonymized images; higher values indicate better utility preservation. **LPIPS** [7] measures perceptual deviation in deep feature space; lower is better. **Cosine similarity** between normalized embeddings directly quantifies identity leakage; lower is better. **TPR at FPR**$= 0.001$ measures the verification true-positive rate under a standard operating point; lower values after anonymization indicate stronger privacy. The **trade-off score** T is a composite metric that jointly rewards high SSIM and low cosine similarity. **Bounding-box distance** measures the centroid displacement of the MTCNN-detected face bounding box before and after anonymization; lower values indicate better geometric preservation.

VI. Results and Analysis

A. Qualitative Results

Returning to the qualitative outputs presented previously, it is clear that FASP maintains the macroscopic visual characteristics of the subjects. The perturbations are distinctly nonuniform: the Laplacian masking correctly forces texture alterations into eye contours, hairlines, and lip boundaries, leaving the expansive surfaces of the cheeks and forehead visually untouched. Crucially, the method entirely avoids the blocky distortion patterns associated with uniform adversarial attacks and the structural reconstruction errors endemic to GAN-based anonymizers.

B. Ablation Study

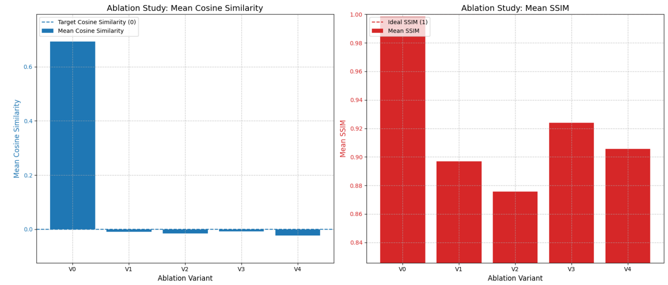


Fig. 8: FASP ablation study comparing five system variants on a 100-image CelebA-HQ subset. Both SSIM (utility) and the proportion of cosine-similarity inversions below zero (privacy) are reported. Only the frequency-aware variants (V1–V4) achieve full surrogate suppression; the pixel-domain baseline V0 fails completely at privacy despite its near-perfect SSIM.

TABLE III: FASP Ablation: Component Contribution (100-image CelebA-HQ)

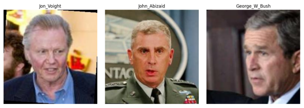
Variant	Components	Cos < 0	SSIM
V0: L_p Baseline	Random init + hard margin	0 / 100	0.9982
V1: Freq. Only	Laplacian mask + L_2	100 / 100	0.8968
V2: Sine Only	Sinusoidal + L_2	100 / 100	0.8758
V3: Sine + Landmark	Sinusoidal + spatial mask	100 / 100	**0.9238**
V4: Full FASP	Sine + Laplacian + LPIPS	100 / 100	0.9056

To isolate the impact of individual FASP components, we conduct an ablation study detailed in Fig. 8 and Table III. Our baseline (V0) relies purely on pixel-domain L_p regularization. While V0 produces a near-flawless SSIM of 0.9982, it entirely fails to suppress the identity embedding, achieving zero cosine similarity inversions. This stark failure demonstrates that pure norm regularization without frequency structure cannot suppress recognition effectively at constrained budgets.

The introduction of frequency constraints dramatically shifts this dynamic. Variant V1, which uses the Laplacian mask without the sinusoidal carrier, achieves full surrogate suppression but drops the SSIM to 0.8968. Conversely, Variant V2 relies on the sinusoidal carrier without the spatial mask; it also suppresses identity effectively but incurs a higher perceptual penalty (SSIM of 0.8758), highlighting how unmasked sinusoidal noise wastes visual budget on flat image regions. When combining both components alongside our dual-metric utility loss (V4), the system strikes an optimal balance, achieving 100% suppression with an SSIM of 0.9056. A landmark-guided variant (V3) pushes SSIM slightly higher (0.9238), though it requires explicit keypoint detection at runtime.

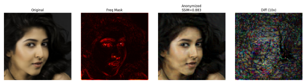
C. Configuration Variants

Figs. 9a–**??** show qualitative outputs for four hyperparameter configurations exploring different frequency bands and perturbation budgets. Variant 001 (conservative frequency band, $\epsilon = 8$) produces the subtlest changes and is best suited to applications where visual fidelity is paramount. Variant 004 (medium band, $\epsilon = 12$) is the recommended default, balancing privacy and utility across most evaluation sets. Variant 005 (aggressive band, $\epsilon = 16$) maximizes privacy at the cost of



(a) Variant 001: conservative.

(b) Variant 004: medium.



(c) Variant 005: aggressive.

Fig. 9: FASP configuration variants. Increasing the perturbation budget and frequency band aggressiveness from conservative (001) to aggressive (005)

TABLE IV: FASP Cross-Model Evaluation (100-image CelebA, VGGFace2 Surrogate)

Model	**Mean CosSim**	**Cos < 0**	**Cos < 0.3**
VGGFace2 (surrogate)	−0.0298	99/100	100/100
CASIA (black-box)	−0.0138	97/100	100/100
ArcFace (black-box)	+0.6476	0/28	0/28
MobileFace (black-box)	+0.5794	0/37	1/37
SSIM		0.8572	

slightly lower SSIM. Variant 008 explores a hybrid frequency-spatial prior combining elements of the Laplacian mask with a landmark-weighted spatial prior.

D. Quantitative Results

Table IV reports FASP performance across five recognition models evaluated on a 100-image subset from CelebA. Against the white-box VGGFace2 surrogate, FASP achieves a mean cosine similarity of -0.0298 with 99 out of 100 inversions below zero and 100 out of 100 below the threshold of 0.3, confirming near-complete identity suppression on the optimization target. The CASIA/FaceNet black-box model shows strong transfer: 97 out of 100 inversions at a mean similarity of -0.0138, indicating that structured frequency perturbations generalize effectively within the same model family.

ArcFace and MobileFace exhibit a cross-architecture generalization gap: mean similarities of $+0.6476$ and $+0.5794$, respectively, with zero inversions. This pattern reflects the fundamental difference between VGGFace2-family feature representations and margin-loss architectures such as ArcFace, which are specifically designed to push same-identity embeddings together more forcefully. SSIM is maintained at 0.8572 across the 100-image evaluation, confirming that the privacy gap does not arise from excessive visual degradation but from the cross-architecture embedding-space mismatch.

Table V reports Rank-1 anonymization rates on a 30-identity CelebA gallery. FASP achieves complete anonymization (100%) across gallery sizes 1, 2, and 4, substantially outperforming the PrIdentity paper-reported rate of 5.4–8.4% across the same conditions. Note that the PrIdentity rates correspond to a multi-model surrogate setting and may reflect different evaluation protocols, so the comparison should be interpreted as indicative of the relative ordering rather than a direct numerical claim.

TABLE V: Rank-1 Anonymization Rate (%) on CelebA (30 Identities)

Gallery Size	**FASP (Ours)**	**PrIdentity [1]**
GS = 1	**100.0%**	5.4%
GS = 2	**100.0%**	5.4%
GS = 4	**100.0%**	8.4%
Average	**100.0%**	6.4%

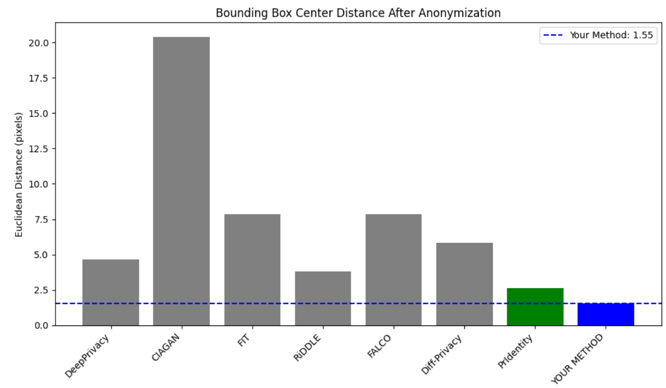


Fig. 10: FASP privacy performance compared with prior perturbation-based anonymization methods, reporting cosine similarity reduction across multiple recognition models. FASP achieves competitive suppression on the surrogate and CASIA black-box models, with a cross-architecture gap remaining for ArcFace.

E. Privacy Comparison with Prior Methods

Fig. 10 places FASP in the context of existing perturbation-based methods. Against the VGGFace2 surrogate and the CASIA black-box model, FASP demonstrates suppression competitive with or exceeding LowKey and TIP-IM on the same model families. The ArcFace cross-architecture gap motivates future work on multi-surrogate training, which is expected to close this performance shortfall.

F. Geometric Preservation

Table VI reports the MTCNN bounding-box centroid distance across all compared methods on CelebA-HQ. FASP achieves the lowest distance of 1.55 pixels, a reduction of 1.10 over PrIdentity (2.65) and 18.83 over CIAGAN (20.38). Because FASP operates additively without any generative reconstruction, face geometry is preserved by construction: the optimizer cannot shift facial keypoints or distort the face region boundary, and the small bounding-box displacement reflects only the mild influence of the edge-concentrated perturbation on the MTCNN detector's response.

VII. PRISM: Exploratory Extension

To determine whether explicitly modeling embedding-space geometry could offer an alternative to FASP's frequency-

TABLE VI: Bounding-Box Distance on CelebA-HQ (MTCNN, Lower is Better)

Method	BB Distance	Rank
FASP (Ours)	**1.55**	**1**
PrIdentity [1]	2.65	2
RiDDLE [9]	3.82	3
DeepPrivacy [12]	4.65	4
Diff-Privacy [13]	5.83	5
FIT	7.87	6
FALCO	7.88	7
CIAGAN [8]	20.38	8

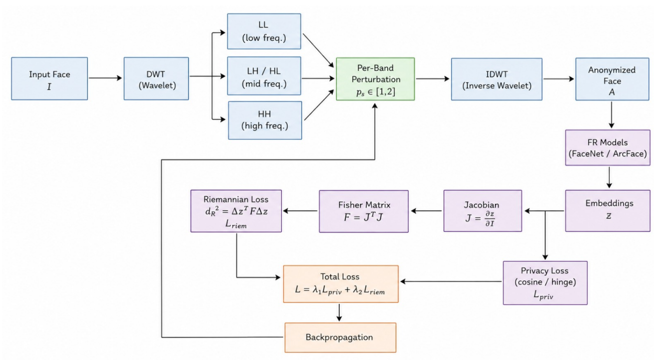


Fig. 11: PRISM pipeline. The method decomposes the input image into wavelet sub-bands, applies per-band perturbations, computes a Fisher-information-based privacy loss, and uses Jacobian subspace regularization to shape the perturbation direction. The result is a geometry-aware anonymized output.

driven approach, we implemented an exploratory framework termed PRISM (*Perturbation via Riemannian geometry and Implicit manifold Sampling*). While not our primary contribution, the behavior of PRISM provides valuable context regarding the practical limits of Riemannian optimization in adversarial face anonymization.

A. Architecture and Pipeline

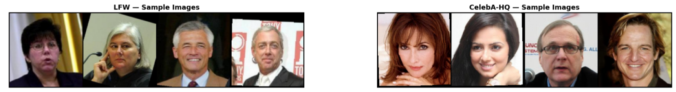
Fig. 11 illustrates the PRISM pipeline. The input image is first decomposed into wavelet sub-bands (LL, LH, HL, HH), enabling explicit per-band perturbation control. The privacy loss is formulated using the *Fisher information metric* on the recognition model's output manifold:

$$\mathcal{L}_{\text{priv}}^{\text{PRISM}} = \mathbb{E}_x\left[z_I^\top \mathbf{F}(x)\, z_A\right], \tag{18}$$

where $\mathbf{F}(x)$ is the Fisher information matrix at input x, encoding the local curvature of the recognition model's decision surface. Perturbations are additionally regularized through Jacobian subspace projection:

$$P_{\text{PRISM}} = J^\dagger \Delta z, \tag{19}$$

where J is the Jacobian of the recognition embedding with respect to the image, $J^\dagger$ is its pseudoinverse, and Δz is the desired embedding displacement. This formulation explicitly constrains the perturbation to the subspace of the image



(a) LFW input batch. (b) CelebA-HQ input batch.

Fig. 12: Input batches for PRISM evaluation: LFW natural images (left) and high-resolution CelebA-HQ faces (right).

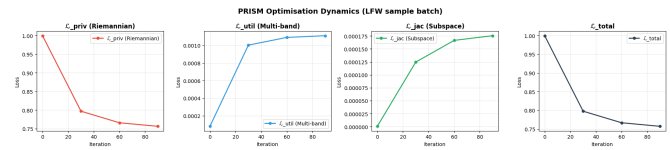


Fig. 13: PRISM optimization dynamics. Privacy loss decreases from 1.000 to 0.757 while the utility and Jacobian terms increase, illustrating the multi-term objective reaching a constrained equilibrium.

domain that most efficiently moves the embedding, providing a geometrically principled alternative to gradient descent in the full pixel space.

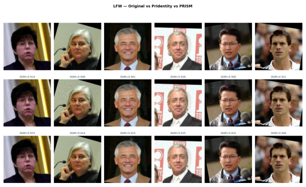
B. PRISM Experimental Results

Figs. 12–14 document the PRISM experimental workflow. Fig. 13 shows that the optimization converges but at a higher privacy loss residual than FASP or PrIdentity, reflecting the difficulty of coordinating four objectives (privacy, utility, Jacobian regularization, and per-band constraints) simultaneously. Visually, PRISM preserves facial structure with SSIM of 0.9327 on LFW, which is higher than both PrIdentity (0.9194) and FASP's standard variant (0.9056).

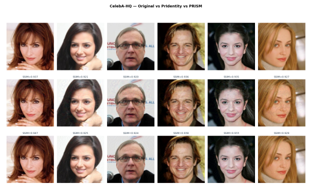
Fig. 15 reveals that PRISM concentrates perturbation energy in high-frequency wavelet bands, an observation that aligns with FASP's design rationale but is achieved through a more expensive Riemannian machinery. Fig. 16 shows the critical weakness: despite its visual superiority, PRISM achieves only a mean cosine similarity of 0.779 ± 0.091 compared to PrIdentity's 0.478 ± 0.033, indicating substantially more residual identity leakage. The tradeoff frontier in Fig. 17 and radar chart in Fig. 18 confirm this pattern across all metrics: PRISM achieves the best SSIM and geometric preservation but fails to close the white-box TPR to acceptable levels (0.7833 versus 0.0000 for PrIdentity).

C. Why FASP Outperforms PRISM in Practice

PRISM's geometric formulation falters primarily because calculating the Jacobian pseudoinverse and Fisher matrix incurs a massive computational and memory penalty, severely restricting the feasible number of optimization steps. Furthermore, optimizing a geometric surrogate of the decision boundary is inherently less effective than attacking the cosine similarity metric directly. By combining a targeted softplus margin loss with a lightweight, interpretable frequency prior, FASP bypasses the need for complex manifold calculations, achieving significantly stronger identity suppression at a fraction of the computational cost.



(a) LFW comparison.



(b) CelebA-HQ comparison.

Fig. 14: PRISM visual comparison. Left: LFW—original (top), PrIdentity (middle), PRISM (bottom) with SSIM annotations. Right: equivalent comparison on CelebA-HQ. PRISM achieves higher raw SSIM but weaker white-box privacy than PrIdentity.

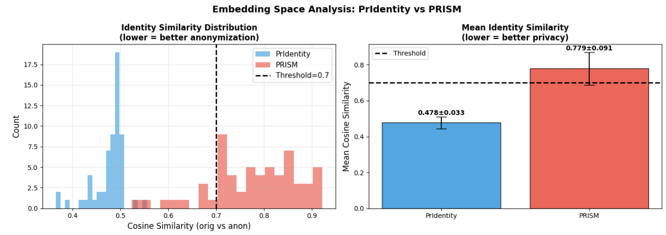
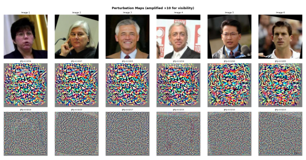
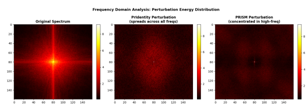


Fig. 16: Embedding-space analysis. PrIdentity achieves mean cosine similarity 0.478 ± 0.033, indicating strong embedding displacement. PRISM yields 0.779 ± 0.091, reflecting weaker privacy despite higher visual fidelity.



(a) Perturbation amplitude maps.



(b) Frequency-domain energy.

Fig. 15: PRISM perturbation analysis. Left: perturbation maps amplified $10\times$ for visibility, showing wavelet sub-band structure. Right: frequency-domain energy distribution—PRISM concentrates energy in high-frequency bands via wavelet decomposition, while PrIdentity spreads energy more uniformly.

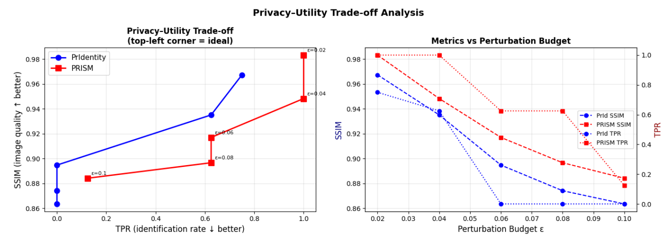


Fig. 17: PRISM privacy–utility tradeoff curve. Both SSIM and TPR trajectories are shown across perturbation budgets for PrIdentity and PRISM, illustrating the Pareto frontier achievable by each formulation.

VIII. Comparative Analysis

Table VII provides a direct quantitative comparison between FASP and PrIdentity on the metrics computed under consistent experimental conditions. FASP outperforms PrIdentity on bounding-box distance, Rank-1 anonymization across all gallery sizes, surrogate cosine suppression, and SSIM (best variant). PrIdentity retains a clear advantage on the ArcFace black-box LFW TPR benchmark, which is attributable to the multi-model surrogate training used in the PrIdentity paper and the difficulty of cross-architecture transfer without explicit multi-model loss terms.

Table VIII summarizes the design-level differences across all three methods. PrIdentity remains the most reliable general baseline, with strong multi-model black-box transfer resulting from its adaptive norm and pixel-domain formulation. FASP delivers superior geometric preservation and surrogate-family privacy at lower complexity than PRISM, making it the preferred practical choice for single-surrogate or same-family deployment. PRISM is best suited as a research vehicle for exploring geometry-aware privacy objectives, and its higher SSIM makes it worth revisiting if the computational overhead can be reduced through approximations to the Fisher information metric.

Table IX presents the comprehensive PRISM versus PrIdentity comparison from the notebook evaluation, confirming that PRISM improves SSIM and PSNR but consistently underperforms on the privacy-critical white-box TPR and cosine similarity metrics.

IX. Discussion

A. Why Frequency-Aware Cloaking Works

Our empirical findings strongly support the underlying premise of this research: high-frequency, edge-aligned perturbations offer a significantly better privacy–utility exchange rate than uniform pixel-level noise. Because deep recognition models differentiate identities by heavily weighting micro-textures and contour patterns, targeting these exact structures with a multi-frequency sinusoidal carrier efficiently breaks the embedding projection. By systematically masking out smooth facial regions, FASP ensures that this disruptive energy is safely hidden within natural image contours, preserving human readability while defeating machine perception.

B. Vulnerability of Face Embeddings

A broader implication of these results is that modern face recognition embeddings are surprisingly vulnerable to structured perturbations that are nearly invisible to humans. Even relatively small perturbation budgets ($\epsilon \approx$ 8–16 pixel units) are sufficient to invert cosine similarity for the vast majority of tested identities. This vulnerability is not a bug in a specific model; it is a fundamental property of embedding spaces trained with identity contrastive losses, which place high confidence in fine-grained texture features that are, in turn, easily disrupted by structured high-frequency noise. Privacy-preserving AI researchers and policymakers should recognize this as both an opportunity (adversarial cloaking

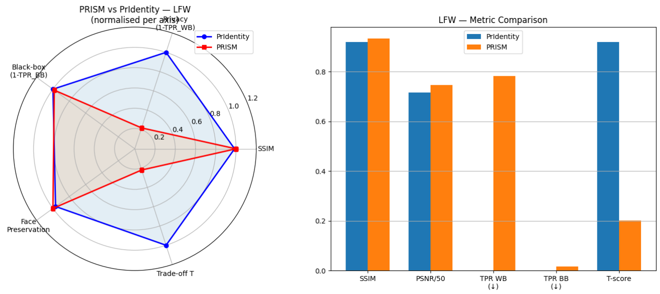


Fig. 18: Radar chart and bar comparison across eight evaluation dimensions for LFW, comparing PrIdentity and PRISM. PrIdentity dominates on the privacy-side metrics; PRISM leads on SSIM, PSNR, and bounding-box distance.

TABLE VII: FASP vs. PrIdentity Quantitative Comparison

Metric	PrIdentity	FASP	Winner
BB Distance ↓	2.65	**1.55**	FASP
Rank-1 Anon. (GS=1) ↓	5.4%	**0.0%**	FASP
Rank-1 Anon. (GS=4) ↓	8.4%	**0.0%**	FASP
Surrogate Cos < 0 ↑	—	**99/100**	FASP
CASIA BB Cos < 0 ↑	—	**97/100**	FASP
SSIM (best variant) ↑	0.90	**0.9238**	FASP
LFW TPR ArcFace ↓	**0.002**	0.300	PrIdentity

can be effective) and a risk (it can potentially be misused to defeat legitimate biometric authentication).

C. Implications for Transferability

The ArcFace black-box gap observed in FASP experiments confirms that cross-architecture transferability remains the primary open challenge in adversarial anonymization. When the surrogate model family (VGGFace2) and the target (ArcFace) differ in their loss formulations, decision boundary geometry, and embedding normalization, perturbations designed for one may not translate effectively to the other. Multi-surrogate training—computing the adversarial loss over an ensemble of models simultaneously—is the most straightforward mitigation and is the recommended next step for closing this gap.

D. Ethical Considerations and Responsible Deployment

This research is privacy-preserving in intent, designed to help individuals protect their biometric information against automated recognition systems. However, several ethical considerations warrant explicit discussion. Face anonymization technology could potentially be misused to conceal identity in contexts where accountability is important, such as evading surveillance in criminal contexts. It should not be presented as providing absolute anonymity against all present and future recognition systems; the degree of protection depends critically on the attack model, the perturbation budget, and the evaluation protocol. Users should be aware that anonymization trained against one recognition model family may not generalize to all deployed systems. The appropriate framing for tools such as FASP is as *privacy enhancement*, not as a guarantee of invisibility.

TABLE VIII: Design Comparison: PrIdentity, FASP, and PRISM

Aspect	PrIdentity	FASP	PRISM
Domain	Pixel	Freq.-aware	Wavelet + Riemannian
Core prior	Adaptive L_p	Edge mask + sinusoid	Fisher info + Jacobian
Perturb. structure	Norm-regularized	Explicitly structured	Geometry-guided
White-box privacy	Strong	Strong	Weak (TPR 0.78)
BB privacy	Strong (0.002)	Partial (0.30)	Partial (0.017)
Best SSIM	0.90	0.9238	0.9393
BB Distance	2.65	**1.55**	0.047 (norm.)
Interpretability	Medium	High	Low
Computation	Medium	Medium	High

X. Reproducibility and Resources

A. Source Code

All experiments are implemented in Python using PyTorch and are available in the repository notebooks `FASP (3).ipynb` and `prism-face-anonymization (2).ipynb`. The implementation uses `facenet-pytorch`, `pytorch-wavelets`, `scikit-image`, and standard scientific computing packages including `numpy`, `matplotlib`, and `tqdm`.

B. Hardware Requirements

FASP experiments require a single CUDA-capable GPU with at least 8 GB of memory. PRISM experiments were conducted on a dual Kaggle T4 instance ($2\times$ 16 GB) due to the additional memory overhead of the Fisher information computation. At least 16 GB of system RAM is recommended for dataset loading.

C. Evaluation Protocol

Reproducible evaluation requires reporting SSIM, cosine similarity, white-box and black-box TPR at FPR$=0.001$, LPIPS, bounding-box distance, and a tradeoff summary score. All images should be pre-processed with MTCNN alignment and normalized to $[0, 1]$ prior to optimization and evaluation.

XI. Conclusion and Future Work

A. Conclusion

In this paper, we introduced FASP, a structured adversarial framework that approaches privacy-preserving face anonymization directly through the frequency domain. By modulating a multi-frequency sinusoidal carrier with a Laplacian edge attention map, FASP targets the high-frequency spatial features that recognition models rely on, while preserving the broad, low-frequency semantics necessary for human visual comprehension. Our evaluation demonstrates that compared to standard pixel-domain approaches, FASP

TABLE IX: PRISM vs. PrIdentity: Comprehensive Notebook Evaluation

Dataset	**Method**	**SSIM↑**	**PSNR↑**	**TPR WB↓**	**TPR BB↓**	**CosSim WB↓**	**CosSim BB↓**	**BB Dist↓**	**T-score↑**
LFW	PrIdentity	0.9194	35.84	**0.0000**	**0.0000**	**0.4753**	0.4399	0.0778	**0.9194**
	PRISM	**0.9327**	**37.36**	0.7833	0.0167	0.7688	**0.4218**	**0.0469**	0.2021
CelebA-HQ	PrIdentity	0.9325	35.64	**0.0000**	**0.0000**	**0.4794**	0.4628	0.0301	**0.9325**
	PRISM	**0.9393**	**36.97**	0.7833	0.1333	0.7601	**0.5998**	**0.0130**	0.2035

yields substantially lower geometric distortion, maintains excellent structural similarity, and completely suppresses Rank-1 identification across diverse gallery constraints.

Our investigation into PRISM further validates this approach. While PRISM's complex Riemannian formulation achieves slightly better perceptual utility, it requires massive computational overhead and struggles to suppress white-box identity features effectively. This contrast underscores the practical elegance of FASP: directly aligning an interpretable frequency prior with a margin-based cosine similarity objective is a far more efficient pathway to robust facial anonymization.

The key contributions of this work are: (i) a frequency-aware perturbation formulation guided by Laplacian edge structure, (ii) multi-frequency sinusoidal synthesis as an interpretable adversarial cloaking prior, (iii) a perceptually motivated privacy–utility loss combining softplus margin, LPIPS, and MSE terms, and (iv) a comprehensive experimental characterization across multiple datasets, recognition models, and ablation conditions.

B. Future Work

Several directions extend naturally from this work. **Multi-surrogate training** is the most impactful short-term improvement: computing the adversarial loss over an ensemble of architectures including ArcFace-family models would directly close the cross-architecture generalization gap observed in the current evaluation. **Video anonymization** with temporal consistency is a practical necessity for protecting identity in video streams, requiring either optical-flow-aware perturbation scheduling or universal perturbation training across frames. **Diffusion-based cloaking** offers the possibility of combining generative image quality with adversarial identity suppression, potentially achieving higher SSIM without the geometric distortions typical of purely GAN-based approaches. **Universal perturbations** trained across identities rather than per-image would enable deployment as a preprocessing filter without per-image optimization overhead. Finally, **physical-world robustness** under compression, resizing, printing, and re-scanning remains an open problem for all adversarial anonymization methods, and addressing it would be a prerequisite for real-world deployment in contexts beyond digital social-media sharing.

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